

FULL STACK DEVELOPMENT WITH MERN PROJECT DOCUMENTATION

1. Introduction

- **Project Title:** Flight finder - Navigating your Air Travel Options
- **Team id:** LTVIP2026TMIDS24635
- **Team Members:**
 1. **Team Leader:** Jannu Shanmukha Veera Brahman
 2. **Team member:** Vandamsetty Bindu Madhuri
 3. **Team member:** Kadali Pavan Ramu
 4. **Team member:** Udara Venkatesh

2. Project Overview

- **Purpose:**

The purpose of the **FlightFinder** project is to address the challenges faced by users in traditional flight booking systems, such as fragmented information, time-consuming comparisons, and lack of transparency in pricing.

The key objectives of this project are:

- To provide a **centralized platform** for flight search, comparison, and booking
- To reduce **user effort and decision-making time** through smart filtering and recommendations
- To enable **flight agencies** to manage and publish flight details efficiently
- To ensure **data authenticity and trust** through admin approval mechanisms
- To deliver a **secure, scalable, and user-friendly** flight booking experience

By achieving these objectives, FlightFinder aims to enhance user satisfaction, improve operational efficiency, and offer a reliable solution for navigating modern air travel options.

- **Features:**

- Search and filter flights
- Book tickets with user login
- Admin panel for managing flights
- Email confirmation
- Booking history

3. Architecture

Frontend:

Built with React.js using components, hooks, and React Router for navigation. UI powered by Material-UI.

Backend:

Developed using Node.js and Express.js. RESTful API handles all CRUD operations related to users, flights, and bookings.

Database:

MongoDB stores user data, flight details, and booking records. Mongoose is used for schema design and data validation.

Frontend (React.js) Components:

- **Search Page:** Handles user inputs such as source, destination, and travel date.
- **Flight List Page:** Displays available and ranked flight results.
- **Booking Page:** Allows users to enter passenger details and confirm bookings.
- **Admin Dashboard:** Enables approval of agencies and flight listings.

Backend (Node.js/Express.js) API Routes: javascript app

```
@app.route('/search', methods=['GET'])
```

```
def search_flights():
```

```
@app.route('/book', methods=['POST'])
```

```
def book_flight():
```

Middleware:

- Authentication and authorization using Flask-Login / JWT
- Input validation and session management

Database (MongoDB) Schemas: javascript

```
const User = new Schema({ email: {  
  type: String, unique: true },  
  bookings: [{ type: Schema.Types.ObjectId, ref: 'Booking' }]  
});
```

4. Setup Instructions

Prerequisites

- Node.js >= 18
- MongoDB installed and running
- npm or yarn Installation
- git clone
<https://github.com/your-username/FlightFinder.git>
- cd FlightFinder
- cd server
- npm install
- cd ../client
- npm install

Environment Variables:

Create a .env file in the /server folder with:

PORT=3000

MONGODB URI-

mongodb+srv://bindu:<db_password>@cluster0.zghkpqk.mongodb.net/?retryWrites=true&w=majority&appName=Cluster0

5. Folder Structure

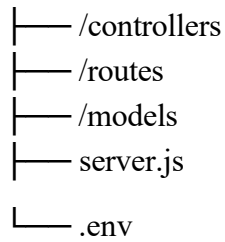
- **Client:**

/client

```
├── /src
|   ├── /components
|   ├── /pages
|   ├── /api
|   ├── App.js
|   └── index.js
```

- **Server:**

/server



6. Running the Application

Frontend:

- cd client
- npm start

Backend:

- cd server
- npm start

7. API Documentation

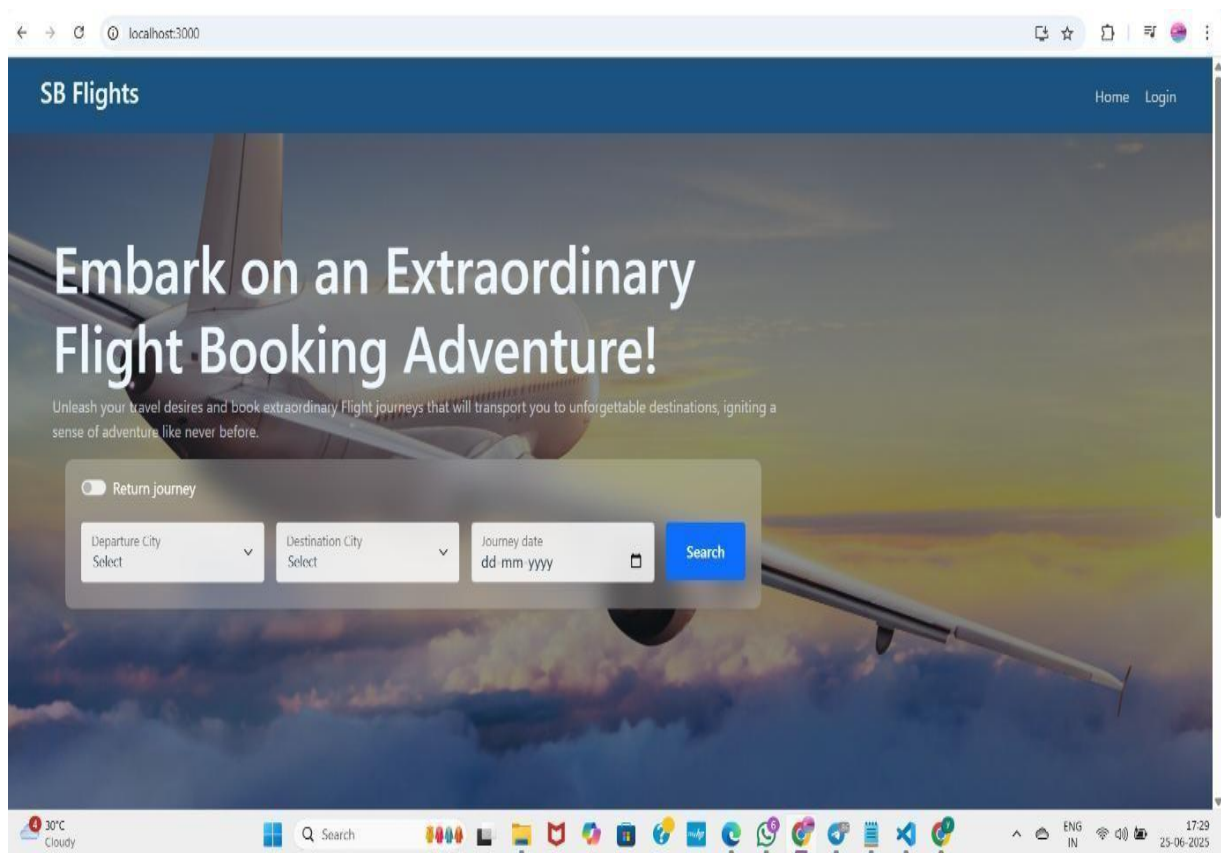
- **POST /api/auth/register**
Registers a new user.
Request Body: { name, email, password }
Response: { success, token }
- **POST /api/auth/login** Logs in
an existing user.
- **GET /api/flights**
Retrieves list of available flights.
- **POST /api/bookings**
Books a flight for the logged-in user.

8. Authentication

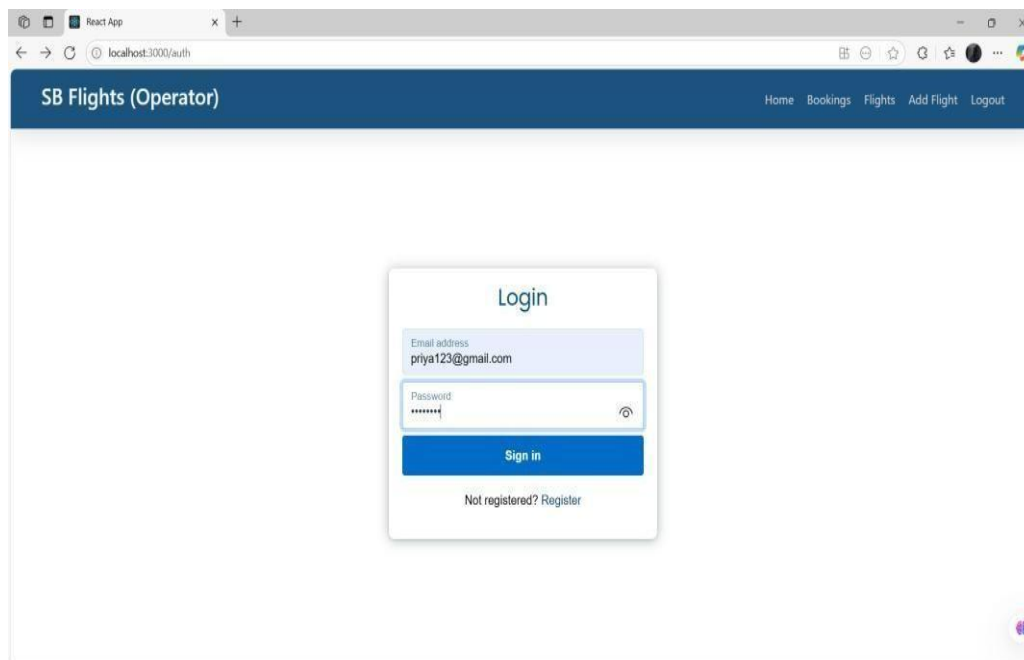
- Implemented using JWT (JSON Web Token).
- Tokens are stored in HTTP-only cookies.
- Protected routes validate tokens before granting access.

9. User Interface

Home Page with flight search



Login form



Register form

The screenshot shows a web browser window with the URL `localhost:3000/auth`. The page has a dark blue header with the text "SB Flights" on the left and "Home Login" on the right. In the center, there is a white "Register" form with a blue border. The form contains the following fields: "Username" with the value "nehapriya", "Email address" with the value "nehhaa12@gmail.com", "Password" with masked characters "*****", and a dropdown menu currently showing "Customer". Below these fields is a blue "Sign up" button. At the bottom of the form, there is a link that says "Already registered? Login".

Admin dashboard

The screenshot shows a web browser window with the URL `localhost:3000/admin`. The page has a dark blue header with the text "SB Flights (Admin)" on the left and "Home Users Bookings Flights Logout" on the right. The main content area features three white cards with blue borders. The first card is titled "Users" and shows a count of "15" with a blue "View all" button below it. The second card is titled "Bookings" and shows a count of "3" with a blue "View all" button below it. The third card is titled "Flights" and shows a count of "3" with a blue "View all" button below it. Below these cards is a section titled "New Operator Applications" which contains a single line of text: "No new requests..".

Booking confirmation page

React App

localhost:3000/book-flight/685bde758b86546e1f4c9d83

SB Flights

Home Bookings Logout

localhost:3000 says
booking successful

OK

Book ticket

Flight Name: priya Flight No: 534
Base price: 9000

Email: sai@gmail.com Mobile: 1234567890

No of passengers: 2 Journey date: 27-06-2025 Seat Class: Economy class

Passenger 1
Name: YARLAGADDA NEHAPRIYA Age: 20

Passenger 2
Name: aarthisri Age: 22

Total price: 18000

Book now

React App

localhost:3000/flight-bookings

SB Flights (Operator)

Home Bookings Flights Add Flight Logout

Bookings

Booking ID: 685cf8c7497807ccff7738a5
Mobile: 1234567890 Email: sai@gmail.com
Flight Id: 534 Flight name: priya
On-boarding: Mumbai Destination: varanasi
Passengers: Seats: E-1, E-2
1. Name: YARLAGADDA NEHAPRIYA, Age: 20
2. Name: aarthisri, Age: 22
Booking date: 2025-06-26 Journey date: 2025-06-27
Journey Time: 10:30 Total price: 18000
Booking status: confirmed

Cancel Ticket

Booking ID: 685be7768b86546e1f4c9ef0
Mobile: 1234567890 Email: sai@gmail.com
Flight Id: 7890 Flight name: priya
On-boarding: Chennai Destination: Pune
Passengers: Seats: E-1, E-2
1. Name: sai, Age: 20
2. Name: ram, Age: 21
Booking date: 2025-06-25 Journey date: 2025-06-28
Journey Time: 03:00 Total price: 16000
Booking status: confirmed

Cancel Ticket

Booking ID: 685bdf958b86546e1f4c9dd4
Mobile: Email:
Flight Id: 534 Flight name: priya
On-boarding: Mumbai Destination: varanasi
Passengers:

10. Testing

- Manual testing on Postman for backend APIs
- Basic component tests in React using Jest and React Testing Library

11. Demo Link:

<https://drive.google.com/file/d/1zip8WSxf9lR-jd4uo2Y-x4vG0q1cmLQm/view>

12. Known Issues

- No password recovery implemented yet
- UI responsiveness needs improvement on tablets

13. Future Enhancements

- Payment integration with Razorpay/Stripe
- Role-based admin access
- Real-time flight tracking integration

